

Sharding

1_Разворачиваем кластер виртуальных машин с помощью файла docker-compose_cluster.yml

```
[yc-user@kafka-node:~]$ sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
24d7e4132556   clickhouse/clickhouse-server        "/entrypoint.sh"        25 seconds ago Up 23 seconds 9009/tcp, 0.0.0.0:9124->8123/tcp, [::]:9124->8123/tcp, 0.0.0.0:8003->9000/tcp, [::]:8003->9000/tcp, 0.0.0.0:9365->9363/tcp, [::]:9365->9363/tcp
1e40b863bb01   clickhouse/clickhouse-server        "/entrypoint.sh"        25 seconds ago Up 23 seconds 9009/tcp, 0.0.0.0:9123->8123/tcp, [::]:9123->8123/tcp, 0.0.0.0:8002->9000/tcp, [::]:8002->9000/tcp, 0.0.0.0:9364->9363/tcp, [::]:9364->9363/tcp
c0f0b0c9b540   clickhouse/clickhouse-server        "/entrypoint.sh"        25 seconds ago Up 23 seconds 9009/tcp, 0.0.0.0:9125->8123/tcp, [::]:9125->8123/tcp, 0.0.0.0:8004->9000/tcp, [::]:8004->9000/tcp, 0.0.0.0:9366->9363/tcp, [::]:9366->9363/tcp
97364acd59ff   confluentinc/cp-kafka:7.8.0        "/etc/confluent/dock... 25 seconds ago Up 23 seconds 0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:9999->9999/tcp, [::]:9999->9999/tcp
d2d812ead29f   clickhouse/clickhouse-server        "/entrypoint.sh"        25 seconds ago Up 23 seconds 9009/tcp, 0.0.0.0:9126->8123/tcp, [::]:9126->8123/tcp, 0.0.0.0:8005->9000/tcp, [::]:8005->9000/tcp, 0.0.0.0:9367->9363/tcp, [::]:9367->9363/tcp
c538a003e54a   grafana/grafana-enterprise:latest   "/run.sh"                25 seconds ago Up 23 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
c063a0305560   zookeeper:latest                   "/docker-entrypoint. ..." 25 seconds ago Up 23 seconds 0.0.0.0:2888->2888/tcp, [::]:2888->2888/tcp, 0.0.0.0:3888->3888/tcp, [::]:3888->3888/tcp, 8080/tcp, 0.0.0.0:2182->2181/tcp, [::]:2182->2181/tcp
8dc33e684b10   prom/prometheus:latest             "/bin/prometheus --c..." 25 seconds ago Up 23 seconds 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp
```

2_Описываем несколько топологий в конфигурации ClickHouse

3_ Предоставить xml

```
SELECT cluster,
        shard_num,
        replica_num,
        host_name
FROM system.clusters
```

```
SELECT
        cluster,
        shard_num,
        replica_num,
        host_name
FROM system.clusters
```

Query id: c8ca382f-2ef1-4209-9bd7-ff3f775f2654

	cluster	shard_num	replica_num	host_name
1.	all_replicated	1	1	clickhouse1
2.	all_replicated	1	2	clickhouse2
3.	all_replicated	1	3	clickhouse3
4.	all_replicated	1	4	clickhouse4
5.	all_sharded	1	1	clickhouse1
6.	all_sharded	2	1	clickhouse2
7.	all_sharded	3	1	clickhouse3
8.	all_sharded	4	1	clickhouse4
9.	default	1	1	localhost
10.	main	1	1	clickhouse1
11.	main	1	2	clickhouse2
12.	main	2	1	clickhouse3
13.	main	2	2	clickhouse4

4_ Создаем distributed таблицу

```

CREATE DATABASE IF NOT EXISTS task_10_distr ON CLUSTER main;
CREATE DATABASE IF NOT EXISTS task_10_as_distr ON CLUSTER all_sharded;
CREATE DATABASE IF NOT EXISTS task_10_ar ON CLUSTER all_replicated;

CREATE TABLE IF NOT EXISTS task_10_distr.one ON CLUSTER main AS system.one
ENGINE = Distributed(main, system, one, rand());

CREATE TABLE IF NOT EXISTS task_10_as_distr.one ON CLUSTER all_sharded AS system.one
ENGINE = Distributed(all_sharded, system, one, rand());

CREATE TABLE IF NOT EXISTS task_10_ar.one ON CLUSTER all_replicated AS system.one
ENGINE = Distributed(all_replicated, system, one, rand());

CREATE DATABASE IF NOT EXISTS task_10_distr ON CLUSTER main

```

Query id: 3cc540e0-62a9-40d1-9eb5-ebaf5f64cf34

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse3	9000	0		2	0
3.	clickhouse1	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.127 sec.

```

CREATE DATABASE IF NOT EXISTS task_10_as_distr ON CLUSTER all_sharded

```

Query id: 99120c99-aa39-446c-8547-bdf0c7ba1a3a

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse3	9000	0		2	0
3.	clickhouse1	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.095 sec.

```

CREATE DATABASE IF NOT EXISTS task_10_ar ON CLUSTER all_replicated

```

Query id: 10586a40-0805-49eb-ae8-f9401eb77b7d

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse1	9000	0		2	0
3.	clickhouse3	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.080 sec.

```

CREATE TABLE IF NOT EXISTS task_10_distr.one ON CLUSTER main AS system.one
ENGINE = Distributed(main, system, one, rand())

```

Query id: 584d6fdf-9555-4c53-9ec8-622d5a3ccf8e

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse3	9000	0		2	0
3.	clickhouse1	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.079 sec.

```
CREATE TABLE IF NOT EXISTS task_10_as_distr.one ON CLUSTER all_sharded AS system.one
ENGINE = Distributed(all_sharded, system, one, rand())
```

Query id: 651c7371-79b5-4a57-a0e1-c441c77e4902

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse1	9000	0		2	0
3.	clickhouse3	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.072 sec.

```
CREATE TABLE IF NOT EXISTS task_10_ar.one ON CLUSTER all_replicated AS system.one
ENGINE = Distributed(all_replicated, system, one, rand())
```

Query id: 4f25ad53-2790-4506-8c06-732167f25a90

	host	port	status	error	num_hosts_remaining	num_hosts_active
1.	clickhouse2	9000	0		3	0
2.	clickhouse1	9000	0		2	0
3.	clickhouse3	9000	0		1	0
4.	clickhouse4	9000	0		0	0

4 rows in set. Elapsed: 0.168 sec.

5_ Предоставить вывод по каждой таблице

```
SELECT *,
    hostname() AS host_name,
    _shard_num AS shard_num
FROM task_10_distr.one;
```

```
SELECT
    *,
    hostname() AS host_name,
    _shard_num AS shard_num
FROM task_10_distr.one
```

Query id: 975fa754-bac2-4cc2-8e4d-e255eb270aa3

	dummy	host_name	shard_num
1.	0	clickhouse1	1
2.	0	clickhouse4	2

2 rows in set. Elapsed: 0.026 sec.

```
SELECT *,
    hostname() AS host_name,
    _shard_num AS shard_num
FROM task_10_as_distr.one;
```

```
SELECT
    *,
    hostname() AS host_name,
    _shard_num AS shard_num
FROM task_10_as_distr.one
```

Query id: 532594ea-062a-4ad8-9d61-d078cc2ef7cd

	dummy	host_name	shard_num
1.	0	clickhouse1	1
2.	0	clickhouse4	4

3.	0	clickhouse3	3
4.	0	clickhouse2	2

4 rows in set. Elapsed: 0.018 sec.

```
SELECT *,
    hostName() AS host_name,
    _shard_num AS shard_num
FROM task_10_distr.one;
```

```
SELECT
    *,
    hostName() AS host_name,
    _shard_num AS shard_num
FROM task_10_distr.one
```

Query id: 48a9ffd3-6655-4833-b383-5ef468e8101d

	dummy	host_name	shard_num
1.	0	clickhouse1	1
2.	0	clickhouse3	2

2 rows in set. Elapsed: 0.011 sec.

Заголовок